# Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the 📝 boxes is helpful for understanding the concept(s).

## 13.1 Conditions for Efficiency

### Marginal benefit and consumer surplus

- Total benefit and marginal benefit

| Total benefit | Marginal benefit |
|---|---|
| It refers to the benefit to consumers derived from consuming certain units of a good. | It refers to the benefit to consumers derived from consuming an additional unit of a good. |

MB of the *n*th unit of a good
= TB of *n* units of a good – TB of (*n* – 1) units of a good

- Marginal benefit and demand
  - Marginal benefit is also the maximum amount that a consumer is willing to pay for an additional unit of the good.
  - A demand curve reflects the marginal benefit of every quantity of the good to consumers.
- Consumer surplus
  - **Consumer surplus** refers to the difference between the maximum amount a consumer is willing to pay and the actual amount he or she pays.

Consumer surplus (CS) = Total benefit (ΣMB) − Total expenditure (P × Q)

  - When the price of a good increases (decreases), consumer surplus will decrease (increase).
  - Consumer surplus is maximised when the price of a good equals the marginal benefit of the last unit of the good consumed.

### Marginal cost and producer surplus

- Total cost and marginal cost

| Total cost | Marginal cost |
|---|---|
| In the short run, it is the sum of total fixed cost and total variable cost. | It refers to the change in total cost resulting from producing an additional unit of output. |

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MC of the *n*th unit of a good = TC (or TVC) of *n* units of a good − TC (or TVC) of (*n* −1) units of a good

- Marginal cost and supply
  - Marginal cost is the minimum amount that a producer is willing to receive for producing an additional unit of the good.
  - A supply curve reflects the marginal cost of every quantity of the good to the producer.
- Producer surplus
  - **Producer surplus** refers to the difference between the minimum amount a producer is willing to receive to produce a good and the amount he or she actually receives.

    Producer surplus (PS) = Total revenue (P × Q) – Total variable cost (ΣMC)

  - When the price of a good increases (decreases), producer surplus will increase (decrease).
  - Producer surplus is maximised when the price of a good equals the marginal cost of the last unit of the good produced.

## Measuring total social surplus

- **Efficiency** is achieved when the allocation of resources maximises total social surplus (TSS).
- There are two methods to measure total social surplus:

  Total social surplus = Consumer surplus + Producer surplus

  Total social surplus = Total benefit – Total variable cost

When total social surplus is maximised, efficiency is achieved, but consumer surplus is not necessarily equal to producer surplus.

## Price mechanism and efficiency

- Efficiency
  - Total social surplus is maximised and efficiency is achieved when:

    Marginal benefit (MB) = Marginal cost (MC)

The problem of scarcity still exists when efficiency is achieved.

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- Inefficiency
  - A **deadweight loss** is **the potential gain in total social surplus that is not captured when marginal benefit is not equal to marginal cost**.

| Under-production | Over-production |
|---|---|
| • MB > MC<br>• quantity of output < equilibrium quantity<br>• there is a deadweight loss<br>• TSS will increase if output increases | • MB < MC<br>• quantity of output > equilibrium quantity<br>• there is a deadweight loss<br>• TSS will increase if output decreases |

## Functions of prices

| Rationing function | Allocative function |
|---|---|
| • **Rationing or distributing goods to users with the highest value for the goods**<br>• Only those who are willing and able to pay the market price, i.e. users with the highest value, can get the goods | • **Allocating resources in markets**<br>• When the price of one good relative to that of another good changes, more resources will be switched from the good with decreased relative price to the good with increased relative price for production |

# 13.2 Deviations from Efficiency

Effects of different kinds of market interventions:

| | Price ceiling | Price floor | Quota | Per-unit tax | Per-unit subsidy |
|---|---|---|---|---|---|
| Market price | ↓ | ↑ | ↑ | ↑ | ↓ |
| Quantity transacted | ↓ | ↓ | ↓ | ↓ | ↑ |
| Total expenditure on the good | ↓ | uncertain | uncertain | uncertain | uncertain |
| CS | uncertain | ↓ | ↓ | ↓ | ↑ |
| PS | ↓ | uncertain | uncertain | ↓ | ↑ |
| TSS | ↓ | | | | |
| Deadweight loss | exists | | | | |

Although a per-unit subsidy leads to increases in both consumer surplus and producer surplus, there is a part of government subsidy that is not captured by the consumers and producers. Therefore, efficiency is not improved.

# 13.3 Divergence between Private and Social Costs (Benefits)

## Negative externality and positive externality

- When a person engages in an activity that affects the well-being of an external party but neither pays nor receives any compensation for that effect, an **externality** arises.

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- Negative externality (Detrimental externality)
  - A **negative externality** (**detrimental externality**) exists when a person's action **imposes costs on other parties**, but the person **does not give any compensation**.
  - With the existence of external cost, there is a divergence between private and social costs.

$$\text{Social cost} = \text{Private cost} + \text{External cost}$$

| | |
|---|---|
| Private cost | the cost borne by the person who takes the action |
| External cost | the cost borne by the affected parties who do not receive any compensation |
| Social cost | the cost borne by society as a whole |

- Positive externality (Beneficial externality)
  - A **positive externality** (**beneficial externality**) exists when a person's action **generates benefits to other parties**, but the person **does not receive any compensation**.
  - With the existence of external benefit, there is a divergence between private and social benefits.

$$\text{Social benefit} = \text{Private benefit} + \text{External benefit}$$

| | |
|---|---|
| Private benefit | the benefit gained by the person who takes the action |
| External benefit | the benefit gained by the affected parties who do not pay for it |
| Social benefit | the benefit received by society as a whole |

## Social optimality and market failure

- Social optimality
  - Social optimality is achieved when:

$$\text{MSB} = \text{MSC}$$

  - When there is no externality:

$$\text{MSB} = \text{MPB}$$
$$\text{MSC} = \text{MPC}$$

  - When externality exists, the output determined in the market (where MPC = MPB) will either be higher or lower than the socially optimal level (where MSC = MSB).

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- Market failure: negative externality and positive externality

| Negative externality | Positive externality |
|---|---|
| - MSC is greater than MPC.<br>- Market equilibrium quantity is greater than socially optimal quantity.<br>- At the market equilibrium, MSC is greater than MSB.<br>- There is over-production/ over-consumption of the good. | - MSB is greater than MPB<br>- Market equilibrium quantity is smaller than socially optimal quantity.<br>- At the market equilibrium, MSB is greater than MSC.<br>- There is under-production/ under-consumption of the good. |

## Government and market solutions

- Government solutions

| Solution | Explanation |
|---|---|
| **Tax** | Levying a tax on parties that generate negative externalities.<br>→ Marginal private cost increases.<br>→ Market equilibrium quantity decreases. |
| **Subsidy** | Providing a subsidy to the parties that generate positive externalities.<br>→ Marginal private cost decreases.<br>→ Market equilibrium quantity increases. |
| **Rules and regulations** | • A government may lessen the problem of negative externalities by:<br>- setting a maximum allowable level of harm for firms<br>- restricting the output of the firms that generate negative externalities<br>- banning the production and consumption of goods that generate negative externalities<br>• A government may require people to consume or produce a certain amount of goods that generate positive externalities. |
| **Government provision** | The government may take up the production of goods and services that generate externalities. |
| **Patents and copyrights** | Patents and copyrights create an incentive for people to take into account the external benefits of their inventions or creations. |

- Market solutions
  - When private and social costs (benefits) diverge, regardless of the initial assignment of property rights, as long as property rights are well-defined and exchanges can be conducted at a negligible cost, market forces will deliver an efficient outcome.

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